Characteristic Line and Gas Puff Analysis Using X Ray Data in the PFRC-2

Dini Kathriarachchi[1,2], Madelyn Smoyer[1,2], Andrew Ryan[1,2]
Liam David[1,2], Samuel Cohen[1,2]
[1]Princeton University
[2]Princeton Plasma Physics Laboratory

October 1, 2025

Abstract

The electron energy distribution function (EEDF) is an important parameter for understanding the behavior of a plasma. Since the EEDF cannot currently be directly measured, the x-ray energy distribution function (XEDF) can be used as a proxy to gain a general idea of what the EEDF might look like. X-ray data was collected from the Princeton Field Revered Configuration over the course of six weeks spanning Jun 23, 2025 to August 1, 2025 using an Amtek 123 Silicon Drift Detector. It was determined that the detected Kα emission is significantly influenced by the geometry of the PFRC-2 as well as variations in RMF power, central cell pressure, and magnetic field flux. Additionally, it was shown that the relative brightness of these carbon, nitrogen, and oxygen peaks can be used to estimate the number density of the respective elements within the plasma. The temporal pattern of the x-rays was also studied, both with and without an additional hydrogen disturbance. The hydrogen disturbance led to a loss in x-ray emission, with Helium plasmas recovering faster than Neon plasmas.

1 Introduction

The Princeton Field-Reversed Configuration (PFRC-2) is a small-scale prototype plasma device designed to form and sustain field-reversed configurations (FRCs) while reducing neutron production. FRCs utilize closed field lines to confine plasma to allow for nuclear fusion. The PFRC-2 device employs 1.8 MHz odd-parity rotating magnetic fields (RMF) for pulses up to ∼300ms to enable electron heating within the plasma. To gain a comprehensive understanding of the electron behaviour, and by extension the plasma behaviour, within the PFRC-2, an electron energy distribution function (EEDF) will need to be constructed. EEDFs within the PFRC-2 are expected to possess a Non-Maxwellian shape consisting of a low-energy, approximately Maxwellian bulk and a non-thermal, high-energy tail [1]. The electrons within the plasma produce X-rays predominantly through Bremsstrahlung and characteristic line emission (most prominently Kα). The former is caused by rapid changes in acceleration of high speed electrons due to interactions with other particles and is responsible for the bulk of the emissions detected. The latter is caused by the de-excitation of electrons from higher to lower energy shells and is responsible for peaks at higher energy values. The distribution of X-ray energies is linked to the distribution of electron energies and can be used in the construction of EEDFs. Therefore, studying the X-ray energy distribution function (XEDF) and temporal profile of X-ray emission is essential for diagnosing the energy distribution and dynamics of high-energy electrons in the plasma.

Emitted X-rays were detected and measured using an Amptek X-123 FAST silicon drift detector (SDD) aligned radial to the central cell chamber of the PFRC-2 to view the plasma center through an adjustable aperture. The measurements from the SDD, in conjunction with other instrumentation, can be used to study the XEDF of the plasma. The energy values of detected x-rays, the count of x-rays in each energy bin and the time during an RMF pulse at which each ray reached the SDD's sensor were all used to inform the conclusions of this paper. Sections (2) and (3) are concerned with the shape of the XEDF as detected by the SSD, specifically the relative sizes of characteristic emission lines from impurities present within the central chamber. These emission lines disrupt the tail of the XEDF's

Bremsstrahlung and studying why they appear and how their presence in the XEDF can be minimized will allow us to construct more accurate EEDFs from the XEDFs. Section (4) focuses on the temporal variation of X-rays, especially how the emissions varied over a single pulse. Instabilities were induced in the plasma by injecting puffs of H2 gas into the central cell chamber and the resulting impacts to temporal X-ray profiles were studied closely to understand how the system responds to such disruptions. In both cases, the X-ray emissions detected by the SDD were utilized to gain insights into the behaviour of the plasma as it exists within the PFRC-2.

2 Kα Emission Peaks vs PFRC-2 Parameters

Occasionally, the fast electrons within the plasma collide with bound electrons. If the free electron has an energy greater than the binding energy of the other electron, it can knock the bound electron free and create a vacancy. An electron from an outer shell will then fill this vacancy and emit what is referred to as a characteristic x-ray. For carbon, nitrogen, and oxygen (CNO), the only possible case is an electron from the L shell filling a vacancy in the K shell, known as Kα emission. X-rays are then emitted at 277eV, 392eV, and 525 eV, corresponding to C, N, and O, respectively. These lines appear as broadened peaks on the x-ray spectrum. The presence of these peaks indicates impurities within the plasma, possibly arising from the walls of the PFRC-2 and atmospheric contamination.

2.1 Characteristic CNO Peaks

CNO peaks were observed at pressures ranging from 0.5 - 2.8 mTorr, RMF powers ranging from 80-165 kW, and magnetic flux density Bz up to 342 G. Peaks were observed in Ar, He, and Ne plasmas, but not H plasmas (Table 1). Within these ranges, there was a window where every single data point taken had CNO peaks: 0.5-2.24mTorr, 90-112 kW, and up to 240 G. Outside of this window, there were times when CNO peaks were visible on certain shots but not on others. The geometry of the PFRC-2 is critical to the observation of x-ray data. Any changes made to the positioning of the aperture can substantially impact the detected data. Figure 1 shows three x-ray spectra taken back-to-back under the same conditions on August 1, 2025. The top spectrum was taken with the same aperture geometry as the first 24 shots of the day. The middle spectrum was taken after the aperture was adjusted slightly to reduce the x-ray count rate and mitigate the chances of pulse pile up. Notably, it was still considered to be on the third aperture. The bottom spectrum was taken with a fully open aperture. While the top and bottom spectra both show clear CNO peaks, they differ in their relative brightness. Additionally, the middle spectra seems to show no CNO peaks at all. Thus, it is clear that even minute changes to the aperture can impact the resulting spectra, and drawing comparisons between shots taken on different days with slightly different aperture conditions must be done sparingly. When attempting to determine how parameters such as center cell pressure, RMF power, and magnetic field affect the presence of CNO peaks, it is necessary to account for any known geometry changes during a day's run and consider the possibility of geometry differences between different days.

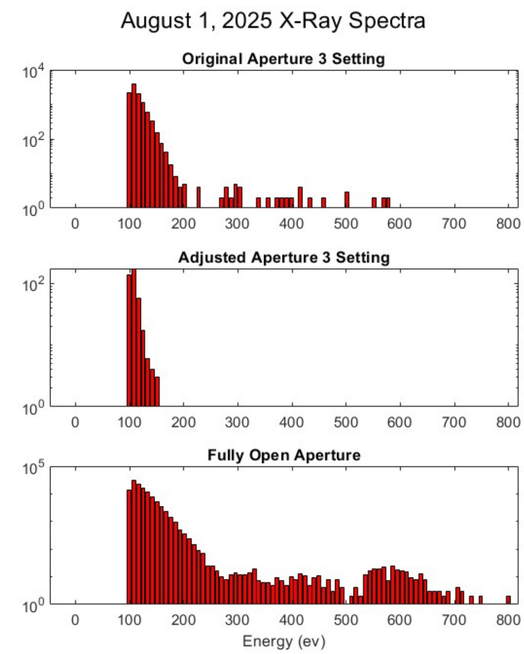

Figure 1: X-Ray Spectra Taken on August 1, 2025, demonstrating the effects of aperture adjustments on detected spectra

Due to a lack of data points on days where Neon, Hydrogen, and Helium were used, it was decided to focus the analysis on argon plasmas only. For the sake of comparison, after all the argon data was collected, the shots that were taken with apertures other than aperture three were eliminated (Figure 2).

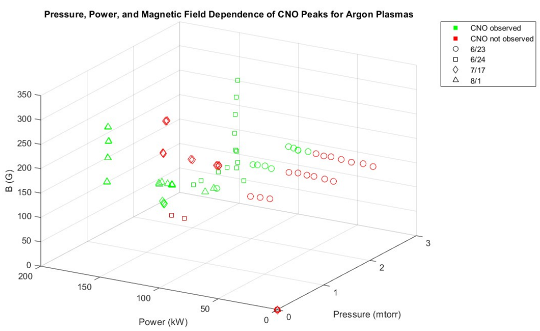

Figure 2: CNO Peak Occurrence for Argon Plasmas at Aperture 3

The geometry differences day-to-day are unknown and cannot be completely accounted for, and, of course, the plasma itself will not be the same and will necessarily have different levels of impurities during each run. For example, CNO peaks were not observed at magnetic fields greater than 176G at low pressure (∼0.5 mTorr) and RMF powers < 120 kW, but the only data points in this range were all taken on 7/17. The only comparable data points were taken on 8/1 at a slightly higher pressure (∼0.56 mTorr) and power (∼122 kW).

2.2 Relative CNO Peak Brightness

An analysis of the relative peak brightness was performed for the CNO peaks. The relative brightness of characteristic lines was determined as the ratio of brightness of the characteristic peak (carbon, nitrogen, or oxygen) to that of the Bremsstrahlung peak, with brightness being the number of x-rays (counts) in a specific energy range detected by the SDD in a time period. Considering a relative value of brightness allows us, to some degree, to account for sudden changes in x-ray emission rates due to instabilities in the system as well as changes in the aperture size of the detector across runs.

The data collected on Aug 1, 2025 was particularly bright in comparison to data collected during all other days of experimentation. It is possible that the plasma from this run was especially impure. Gaseous particles, likely water, being knocked off the walls of the PFRC-2 towards the beginning of the run may have caused a notable increase in central cell pressure, leading to large amounts of oxygen being introduced into the system. Additionally, characteristic neon x-rays were observed on certain runs, indicating that leftover neon from previous runs was contaminating the system. Analyzing patterns in characteristic peak brightness can allow for the identification of impurities within the PFRC-2, their sources, and the parameter values for which their effects are amplified. This will aid in the modification of experimental equipment, procedures, and run plans to minimize (or maximize if calibrating energy scales) the appearance of these impurities in emission spectra and within the system.

2.2.1 Parameter Sweeps

Pressure, power, and magnetic field density sweeps in which each parameter was varied individually were conducted over several days of experimentation. The relative brightness of CNO peaks during these sweeps showed interesting trends.

In the magnetic field sweeps performed on Jun 24, 2025 and August 1, 2025, there was an exponential increase in brightness as magnetic flux density decreased (until ∼100G) as shown in Figure 3.

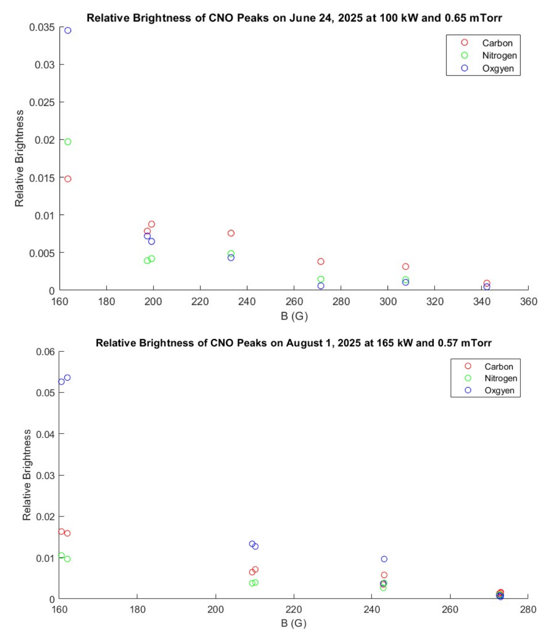

Figure 3: CNO Brightness with Magnetic Field Sweeps

However, the two sets of data contrasted in that the B field sweep on August 1 showed the highest proportions of oxygen, followed by carbon, and then nitrogen for all spectra in which these peaks were visible, while the sweep on June 24 showed that the brightness of oxygen drop significantly more than the brightness of carbon and nitrogen for increasing B values. At flux densities exceeding 230G, oxygen displayed the lowest relative brightness of the three elements.

The RMF power sweeps performed on June 23 and August 1 provided inconclusive results, largely due to a lack of data points. Based on the August 1 data, it appears that increasing RMF power leads to increased brightness (Figure 4). The effect was most pronounced for oxygen and least pronounced for nitrogen. The two sweeps carried out on June 23 explored powers between 80 and 100 kW, a range that was not sufficiently explored on August 1. Both these June 23 sweeps show a dip in brightness at RMF powers of 90 kW, but these sweeps were at a lower magnetic field and higher pressure than the sweep on August 1.

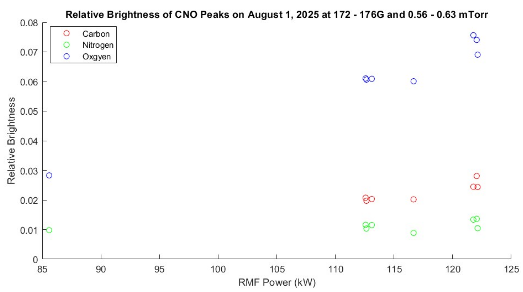

Figure 4: CNO Brightness with RMF Power

A pressure sweep was performed on June 24, 2025. The brightest peaks occurred at 1.3 mTorr. The oxygen peaks were consistently the brightest, followed by carbon and nitrogen (Figure 5).

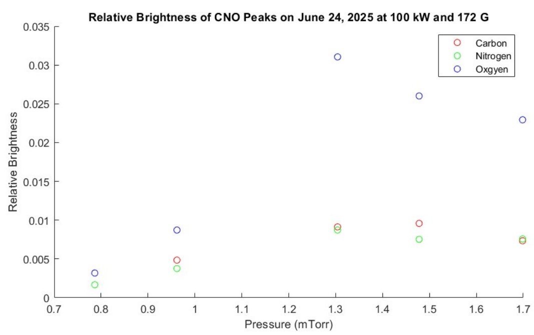

Figure 5: CNO Brightness with Pressure [0.3cm]

2.2.2 Brightness maxima for Power & Pressure

Across all the runs performed during the 6 days of experimentation, 98 runs were selected with magnetic flux densities between 96 G and 126 G and varying central cell (CC) pressures and RMF powers. The magnetic flux density was restricted to this narrow range to reduce the effects of magnetic field on brightness so that the variation of brightness with pressure and RMF power can be investigated more accurately. Figure 6 shows the relative brightness variation with CC pressure and RMF power for carbon (left) and oxygen (right) with biharmonic interpolation used in the generation of the smooth surface plots shown in.

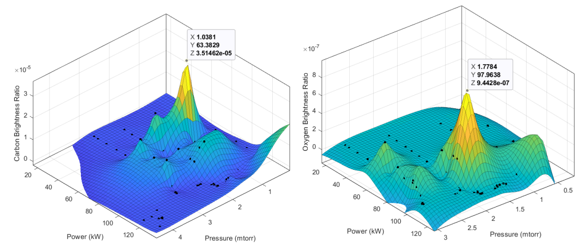

Figure 6: Variation of Relative Brightness of Carbon (left) and Oxygen (right) with Pressure and Power

Carbon brightness appeared to peak at 1 mTorr pressure and 63 kW RMF power, while Oxygen brightness peaked at 1.8 mTorr pressure and 98 kW RMF power. Both elements displayed a slow decrease in brightness as pressure and power increased beyond these values and an abrupt (steeper) decrease as pressure and power reduced below these values. The nitrogen brightness results were not included as they did not exhibit discernible peaks for power and pressure, beyond a general trend of increasing brightness with RMF power. Further investigations is required to better understand relative brightness trends in nitrogen. The results from carbon and oxygen show that characteristic emissions from trace amounts of these elements inside the PFRC-2 appear increasingly (brighter) as RMF power and CC pressure approach their respective peak values.

2.2.3 CNO Brightness Conclusions

There is not enough consistent data to offer any definite conclusions regarding CNO peaks, as there are so many variables at play. Based on the data that was collected this summer, we would expect the brightest CNO peaks to occur at low magnetic fields, high RMF powers, and pressures of around 1.3 mTorr.

The low magnetic field would create poor confinement, increasing the likelihood of fast electrons hitting the walls of the central cell and introducing impurities into the system. The high RMF power would increase the average energy of electrons within the system and thereby make it more likely that a free electron colliding with a bound electron would have the required energy to create a vacancy. The pressure curve is the most interesting of the trends, as it is not inherently obvious why the peak brightness might be at the specific value of 1.3 mTorr. At lower pressures, it is more likely for there to be impurities in the system, so one might expect the brightest peaks to be at the lower pressure values. Conversely, at higher pressures, the likelihood of collisions increases due to an increase in either temperature or number density, so it is also reasonable to expect bright peaks as pressure increases. It is possible that the peak in the center of the pressure range simply offers the optimal combination of collisions and impurities resulting in maximum brightness. More testing needs to be done in order to see if this pattern is replicated at other magnetic fields and powers.

3 CNO Number Density Calculations

The number density of the trace amounts of carbon, nitrogen and oxygen within the system was calculated as follows:

$$\mathrm{n}_{C,N,O} = \frac{C_{tot,rec}}{t_{acc}\,\Omega\,V_{eff}\,n_e\,v_e < \sigma_{CNO} >}$$

where $C_{tot,rec}$ is the total number of counts recorded within the relevant range for the characteristic line, σ is the absolute k-shell ionization cross-section value, Ω is the ratio of solid angle visible to the detector, and V_{eff} is the effective volume as seen by the detector. The relative number density was defined as the ratio of the calculated carbon, nitrogen or oxygen number density to the target particle number density (determined to be the sum of electron density and neutral gas density) within the system.

A hypothesis was put forward suggesting a possible relationship between the relative number density of the three elements and their relative peak brightness in emission spectra. If we assumed all Bremsstrahlung radiation is from the target particles (not necessarily true as fast electrons also contribute) namely plasma ions and neutral gases, then the Bremsstrahlung peak height (brightness) would be proportional to the density of these target particles within the system. Similarly, if we assume the characteristic line emissions were produced solely by Kα emissions from the corresponding elements, then the brightness of those peaks as seen on the emission spectra should be dependent on the number density of these elements inside the PFRC-2. Therefore, such a relationship would be within the bounds of theoretical reasoning.

In consideration of this, the relative brightness and relative number density values were computed for carbon, nitrogen and oxygen for runs involving Helium and Neon plasmas and plotted on a log-log plot to span the entire range of values for both variables [Figure 7]. As predicted, it can be observed that these two ratios are closely related to each other as they are similar in value and follow a general trend of proportionality. In terms of magnitude, the two variables appear to differ, on average, by a factor of $log(0.28) \simeq 0.5$. The Helium data points deviate more from this trend. A possible reason may be that the peaks observed in the emission spectra when Helium plasma was used were small (less counts), leading to greater uncertainty. Overall, as expected, there is a visible trend between relative brightness and relative number density. Given this, it is possible to use the relative brightness as a means to estimate the number density of CNO impurities present within the PFRC-2.

Figure 7: Calculated relative C, N or O number density vs relative brightness for He and Ne plasma runs

4 X Rays Over Time

Examination of the X-ray count as a function of gas puff duration provides important insight into plasma stability and electron energy in the PFRC-2. The PFRC-2 sustains electron heating through the application of an odd-parity rotating magnetic field. By monitoring the temporal evolution of X-ray counts, one can directly assess whether electron heating is sustained under RMF_0 drive and whether the plasma remains on a trajectory toward the temperatures required for effective ion heating and eventual fusion conditions.

4.1 Temporal Profile

Examining the temporal profile of the X-ray counts over the duration of the puff revealed an exponential trend for He and Ne run data. These runs were fit to the function

$$Counts = Ae^{t/\tau} + B \tag{1}$$

as seen in [Figure 8].

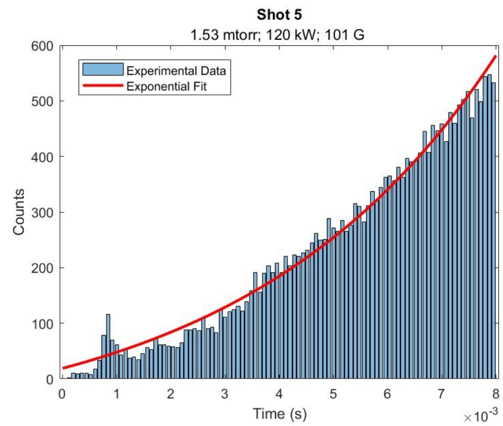

Figure 8: Exponential fit to the X-ray count rate in 0.1 ms bins over an 8 ms He pulse

All He and Ne runs fit this trend with low variance. This is beneficial because it shows RF energy deposition into the electron population outpaces collisional and radiative losses, placing the plasma on a favorable trajectory for increasing ion heating and approaching fusion-relevant temperatures. These runs had a gas pressure from 0.95 – 2.95 mTorr, Bz value 100 G, and 110 – 130 kW RMF power. There is a notable peak at 1ms corresponding to a sharp increase in electron density that was a function of the electrical startup for the PFRC-2. The PFRC-2 tended to overshoot its initial step up to current. Plotting the fit, specifically the parameter versus gas pressure as seen in [Figure 9] reveals a linear trend, indicating that the increase in x-rays rises quicker at lower gas pressures. This dependency may be characteristic of the lower electron population in lower gas pressure systems as the energy transferred to the electrons is not split among a larger population, leading to higher individual electron energies. For future operation, it may be beneficial to operate at lower pressures to rapidly raise electron energy, however, this risks reducing the number density of the ions, a key variable in fusion power output.

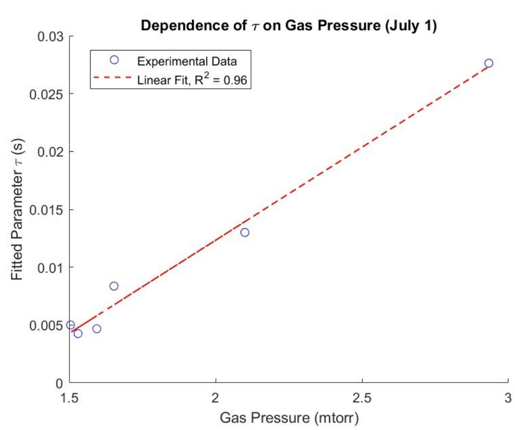

Figure 9: Linear fit to τ parameter versus gas pressure for an 8ms He pulse at 100 G and 120 kW RMF power

4.2 Hydrogen Puff Trends

Additional runs were done that puff in various amounts of hydrogen gas. Puffing hydrogen gas into the PFRC-2 serves as a controlled method to probe plasma stability under varying fueling conditions. Introducing additional neutral gas modifies the plasma density, collisionality, and pressure balance, which in turn tests the resilience of the PFRC-2 to perturbations. Ionizing this hydrogen gas saps energy from the electrons in the system, causing a dip in the X-rays as seen in [Figure 10]. A piecewise fit was applied to this data, and an arbitrarily chosen 12-6 function was added to the exponential

$$D\left[\left(\frac{\sigma}{t-a}\right)^{12} - \left(\frac{\sigma}{t-a}\right)^{6}\right]\{t > \sigma + a\} \qquad (2)$$

to provide a method for quantifying the reduction in counts and recovery time of the puff. As seen in [Figure 10] the X ray counts return incredibly quickly to the previous exponential trend after dipping. This is indicative that the PFRC-2 will be resilient to perturbation and can return to previous trends or behavior rapidly.

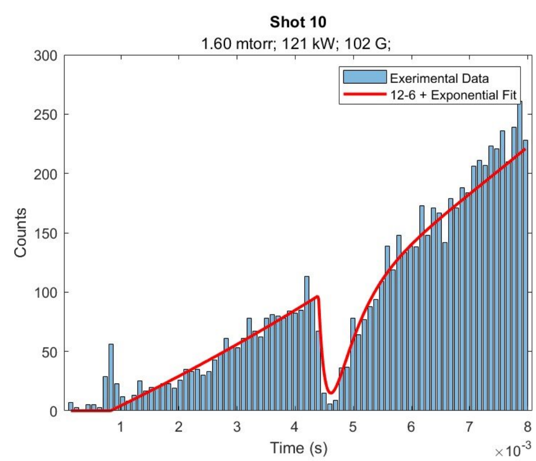

Figure 10: Exponential plus 12-6 fit to the x-ray count rate in 0.1ms bins over 8ms He pulse with a 0.2ms H puff

Plotting both the normalized drop as defined by the fit drop divided by the counts immediately before the drop and the recovery time as the time taken to return to 95 percent of the expected value without the drop yields [Figure 11] and [Figure 12].

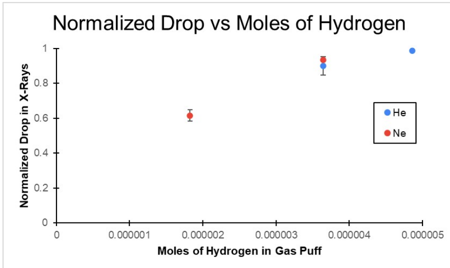

Figure 11: Normalized drop in X-ray counts versus moles of hydrogen in gas puff

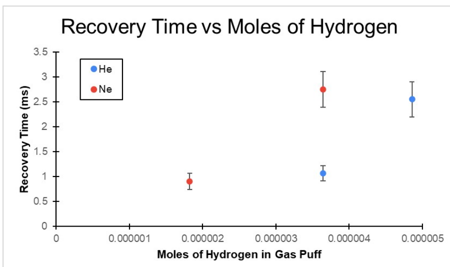

Figure 12: Recovery time in X-ray counts versus moles of hydrogen in gas puff

As expected in [Figure 11] there is a greater drop with more moles of hydrogen puffed and a similar drop seen between gasses since the main effect of more hydrogen puffed in is a greater amount of energy used to ionize that hydrogen. Interestingly, in [Figure 12] while the expected trend of a higher recovery time with more moles of hydrogen puffed appears, Ne generally has higher recovery time. This could potentially be caused by the higher atomic weight of Ne compared to He, decreasing its velocity and therefore rate of collisions that return the energies to their previous state. However, this is unlikely to impact the electron energies, and further study is required.

5 Conclusion

X-ray measurements in the PFRC-2 show that the presence and brightness of carbon, nitrogen, and oxygen peaks depend on several factors, including central cell pressure, RMF power, magnetic field, and even small changes in detector geometry. The brightest peaks generally appeared at low magnetic fields, moderate to high RMF powers, and pressures around 1.3 mTorr, which suggests a combination of electron energy, collisions, and impurities from the walls contribute to what we see in the spectra.

Looking at the relative brightness alongside the calculated number densities, there's a clear trend showing that peak brightness can give a rough estimate of impurity levels in the plasma. The temporal X-ray data also shows that He and Ne plasmas follow a mostly exponential rise during pulses, with lower pressures leading to faster increases. Hydrogen puffs caused noticeable drops in X-ray counts, but the system recovered quickly, which indicates that the plasma can respond to disturbances without long-term effects on electron energy.

This data shows how sensitive X-ray measurements are to both the plasma and the experimental setup. These results give a better picture of how impurities behave, how electrons are heated, and how the PFRC-2 responds to changes, which will help guide future experiments and improve our understanding of the plasma.

6 Acknowledgments

This work is supported by the US DOE Contract No. DE-AC02-09CH11466, Program in Plasma Science and Technology at Princeton University (PPST), and High Meadows Environmental Institute (HMEI). Special thanks to our mentors Prof. Samuel Cohen and Liam David for their guidance and support throughout this research.

7 References

[1] Swanson, C., Jandovitz, P., Bosh, A., Glasser, A., & Cohen, S. (2016). *Extracting Electron Energy Distribution from PFRC X Ray Spectra*

[2] *Journal of Physical and Chemical Reference Data* 43, 013102 (2014); DOI: 10.1063/1.4832851

[3] Thompson, A. C., et al. (2001). *X-Ray data booklet* Pg 35

8 Appendix

Gas	Date	Pressure (mTorr)	RMF Power (kW)	Magnetic Field (G)	CNO Observed
		0	0	0	No
		2.6999	25.02071781	145.8536041	No
		2.6999	33.42245762	148.0484885	No
		2.702	43.39867966	146.4501913	No
		2.7042	52.08481882	148.1181583	No
		2.7092	60.89910725	149.5338837	No
		2.7153	67.25051047	147.9315636	No
		2.7211	74.02503948	149.3204666	No
		2.7327	81.22007805	149.7688378	Yes
		2.7447	90.31062432	148.2405774	Yes
		2.749	90.28699185	147.4537807	Yes
		2.7562	94.39299672	149.8714822	Yes
		2.791	99.99572163	148.639308	Yes
		2.0397	100.5607508	149.4929674	Yes
	6/23	2.0302	96.4066648	151.2218819	Yes
		2.0292	89.97959968	152.1250272	Yes
		2.0209	84.29993102	149.5839284	Yes
		0	0	0	No
		2.0029	68.76740451	149.5912188	No
		1.9937	60.7423746	152.4603948	No
		1.9828	52.75549247	152.10909	No
		1.9674	46.19195306	153.2805419	No
		1.9565	36.7345537	152.6839813	No
		1.9438	28.81831762	152.1347536	No
Argon		0	0	0	No
		1.0415	47.13751371	153.8317575	No
		1.0405	55.31268175	153.0066013	No
		1.0381	63.38288096	151.579418	No
		0	0	0	No
		1.0397	91.89053456	155.2023712	Yes
		0	0	0	No
		0.2935	90.10269236	132.2958299	No
		0	0	0	No
		0.2968	100.7765101	133.8836583	No
		1.7784	97.96384043	130.6904254	Yes
		1.6496	98.78587249	163.5827462	Yes
		1.6526	99.06702261	197.4873546	Yes
		1.6528	100.0225478	199.2806576	Yes
	6/24	1.6543	100.7120931	233.1509157	Yes
		1.6589	100.5927259	271.4357382	Yes
		1.6626	100.097927	307.5750022	Yes
		1.6664	98.945144	342.1938863	Yes
		1.6985	100.0654723	171.642717	Yes
		1.478	100.3449563	172.7726174	Yes
		1.3042	100.9550685	171.8229655	Yes
		0.9617	102.0245928	170.8541508	Yes
		0.7865	101.8691225	171.438295	Yes
		0.7825	102.1347751	172.2584128	Yes
		0	0	0	No
		2.9247	107.9540861	138.0206819	No
		0	0	0	No
		2.9329	108.3501091	151.085275	No
		2.9383	109.5940962	150.549023	No
		2.948	108.6180246	150.53303	No
		0	0	0	No
		1.5266	119.5812326	135.3235111	Inconclusive
		1.5306	120.8007084	133.4523576	Inconclusive
		1.5308	120.7465873	134.1613992	Inconclusive
		1.6392	120.4239098	137.721382	No
		1.6532	121.6956824	135.1727751	No
Helium	7/1	1.6006	121.3192742	136.1075663	No
		0	0	0	No
		1.5946	121.3325184	146.2729229	No
		1.5986	121.2091767	146.3759217	No
		1.5864	120.5767693	137.6613027	Inconclusive
		1.5855	120.8458809	136.3727421	Inconclusive
		1.5041	120.4106962	135.7613675	No
		1.5035	120.4247752	135.6425285	Inconclusive
		1.505	120.6306248	135.9666871	Inconclusive
		1.5085	120.7889692	138.0798867	Inconclusive
		1.507	119.9146337	138.4600716	Inconclusive
		1.5074	120.2522181	139.2647748	Inconclusive
		1.6908	119.8282252	144.1682067	No
		0	0	0	No
		1.4709	121.4948658	146.4421712	No
		1.7081	121.9873476	145.334988	No
		3.9356	124.7672069	131.5508057	No
		0	0	0	No
Hydrogen	7/3	3.8165	116.0075439	135.8021416	No
		4.111	119.4482996	135.4267337	No
		4.3513	118.8610861	135.7938565	No
		4.0347	124.2283171	147.2669862	No
		4.2518	131.6524477	138.107871	No
		3.9697	124.7929574	149.1351668	No

Gas	Date	Pressure (mTorr	RMF Power (kW	Magnetic Field (	CNO Observed
		0	0	0	No
		0	0	0	No
		0	0	0	No
		2.2424	112.686278	142.0014999	Yes
		2.1805	108.4738492	145.281401	Yes
		2.1097	109.8595454	145.7475036	Yes
		2.0989	110.9055894	145.3379475	Inconclusive
		2.0134	119.3509067	150.9716256	Inconclusive
		2.0096	119.1469522	151.4602915	Inconclusive
		2.0193	119.5622453	151.8304431	Inconclusive
Neon	7/8	2.0129	121.5655247	156.5318561	No
		1.9931	121.6981237	156.4510532	No
		1.9689	121.6434076	155.8194043	No
		1.9941	121.7659553	156.4290977	No
		1.9897	117.8659762	144.3942524	No
		1.9821	117.0747306	223.682555	No
		1.9713	116.8674485	223.5549957	No
		1.9752	117.3821569	223.7779979	No
		1.9728	117.9835588	223.2269601	No
		1.9739	117.2074802	223.4487655	No
		1.9781	117.2309143	223.350274	No
		0	0	0	No
		0	0	0	No
		0	0	0	No
		0.4743	40.75271903	131.3711719	No
		0.4783	56.91601019	134.3264472	No
		0.4791	56.74129196	134.716483	No
		0.4809	56.82828788	133.6002731	No
		0.4938	57.58103739	133.8334838	No
		0.5071	82.53612658	138.3374251	No
		0.5142	83.5971428	137.599903	No
		0.5343	84.63468936	136.0495816	Yes
		0.5461	85.97173139	138.4728247	Yes
		0.5016	115.1841521	141.1527374	Yes
		0.4976	115.4984693	141.6522902	Yes
		0.5081	115.6413839	139.3825564	Yes
		0.4993	116.6650794	145.9256483	Yes
	7/17	0.4998	70.12481554	239.8331284	No
		0.5004	70.26681695	239.7586891	No
		0.4969	68.85428951	239.2430088	No
		0.4955	68.93152203	240.5727884	No
		0.4974	70.88731152	239.0224787	No
		0.4982	91.32813973	241.3455017	No
		0	0	0	No
		0.5021	91.48964163	241.2945151	No
		0.506	92.6544471	241.4583258	No
		0.5073	116.3438608	244.4237432	No
		0.5132	116.517939	241.6365093	No
		0.5214	116.5192738	241.9669098	No
		0.5265	117.1406719	241.7838865	No
		0.5304	115.0753072	308.7978816	No
Argon		0.5369	114.380609	309.2893486	No
		0.5372	114.3432211	309.7362249	No
		0.5425	114.683108	308.6199513	No
		0.7647	83.53491256	172.7351413	Yes
		0.6319	85.59045026	171.8856025	Yes
		0	0	0	No
		0	0	0	No
		0.6065	112.5775116	176.0126987	Yes
		0.6072	113.1057273	174.9171533	Yes
		0.6028	112.6490892	174.4553172	Yes
		0.6144	116.6676971	175.1391873	Yes
		0.6207	85.50917743	175.6685044	Yes
		0	0	0	No
		0.6145	121.7680621	176.0539612	Yes
		0.5694	122.0506665	174.3403176	Yes
		0.5646	122.1283657	176.3952544	Yes
		0	0	0	No
	8/1	0.5775	165	242.9080557	Yes
		0.5621	165	243.80691	No
		0.5667	165	243.1539185	Yes
		0.5543	165	210.1774706	Yes
		0.5578	165	209.385009	Yes
		0.5398	165	160.5603397	Yes
		0.5358	165	162.0882801	Yes
		0.5582	165	272.9472329	Yes
		0.5587	165	273.0065831	Yes
		0.5635	165	272.59459	Yes
		0.5615	165	273.9521007	No
		0.5629	165	273.6637585	Yes
		0.563	165	274.2275092	No
		0.5642	165	274.5442918	No
		0.5668	165	275.0200422	No

Table 1: All PFRC-2 Data 6/23 - 8/1